Yibo Zhao

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Education

East China Normal University Sep. 2024 – Present
Master of Engineering in Big Data Technology and Engineering (in progress)
School of Data Science and Engineering. Expected graduation: June 2027.
Advisor: Prof. Xiang Li, PLANING Lab.

East China Normal University Sep. 2020 – June 2024
Bachelor of Arts and Bachelor of Engineering (dual degree)
Journalism and Data Science, School of Communication.

Research Interests

I study how language models and agents acquire information, use evidence, and learn from experience. Technically, I am interested in graph-based retrieval, reinforcement learning for search agents, long-term agent memory, and selective supervision for knowledge distillation.

Publications

* Equal contribution; my name is in bold.

[EMNLP 2026 Main] MERIT: Matching Expertise via Rubric-Informed Training for Reviewer Assignment
Zixuan Yang, **Yibo Zhao**, Weicong Liu, Xiang Li.

[EMNLP 2026 Findings] Skill0.5: Joint Skill Internalization and Utilization for Out-of-Distribution Generalization in Agentic Reinforcement Learning
Jiapeng Zhu, Jianxiang Yu, **Yibo Zhao**, Chengcheng Han, Qi Gu, Xunliang Cai, Xiang Li, Weining Qian.

[EMNLP 2026 Findings] APEX: Academic Poster Editing Agentic Expert
Chengxin Shi*, Qinnan Cai*, Zeyuan Chen*, Long Zeng*, **Yibo Zhao**, Jing Yu, Jianxiang Yu, Xiang Li.

[ACL 2026 Main] RATE: Reviewer Profiling and Annotation-free Training for Expertise Ranking in Peer Review Systems
Weicong Liu*, Zixuan Yang*, **Yibo Zhao**, Xiang Li.

[AAAI 2026] Human Cognition Inspired RAG with Knowledge Graph for Complex Problem Solving
Yao Cheng, **Yibo Zhao**, Jiapeng Zhu, Yao Liu, Xing Sun, Xiang Li.

[EMNLP 2025 Main] Text Detoxification: Data Efficiency, Semantic Preservation and Model Generalization
Jing Yu*, **Yibo Zhao***, Jiapeng Zhu, Wenming Shao, Bo Pang, Zhao Zhang, Xiang Li. *Co-first author.*

[ACL 2025 Findings] Enhancing LLM-based Hatred and Toxicity Detection with Meta-Toxic Knowledge Graph
Yibo Zhao, Jiapeng Zhu, Can Xu, Yao Liu, Xiang Li. *First author.*

Selected Research Contributions

Project titles link to papers; the full publication list is available on my website.

1. Efficient Knowledge Retrieval

E^{2}GraphRAG First author; preprint

Graph-based RAG incurs costly indexing and retrieval. I implemented a summary tree, lightweight entity extraction, and adaptive retrieval, and led evaluation. E^{2}GraphRAG achieved up to 10-fold faster indexing than GraphRAG and 100-fold faster retrieval than LightRAG while maintaining competitive QA performance.

CrossAug Coauthor; under review at ICLR 2027

Chunk-local extraction misses cross-passage relations. I co-developed CrossAug and advised experimental design; targeted graph augmentation improved question answering across three GraphRAG frameworks and four benchmarks.

MetaTox First author; ACL 2025 Findings

Missing domain knowledge and oversensitivity hinder toxicity detection. I proposed and implemented knowledge-graph retrieval; MetaTox reduced false positives while improving detection performance across multiple datasets.

2. Learning and Evaluating Search Agents

Search-Agent Training: A Controlled Study First author; under review at KDD 2027

Confounded evaluations obscure what improves search-agent training. I initiated and led controlled comparisons of retrieval corpora, rewards, and training protocols. Repairing corpus coverage yielded larger gains than algorithm differences; simple outcome rewards remained competitive with or superior to process rewards.

Think Big, Search Small Co-first author; under review at KDD 2027

Uniform model sizing overlooks role-specific capacity needs in search agents. I framed the question and designed controlled experiments. Delegation proved the bottleneck; a distilled 1.7B executor matched a frontier sub-agent using 37% fewer sub-agent tokens.

GRACE First author; under review at ARR

RAG systems often answer without sufficient evidence. I designed a data generation pipeline and gated rewards, integrating training into verl. GRACE improved grounded answering and abstention across two benchmarks using 10% of prior annotation costs.

3. Long-Term Memory and Selective Learning

LazyMem Coauthor; under review at KDD 2027

Early memory compression loses useful details. I advised query-time memory construction and evaluation; LazyMem improved LongMemEval accuracy with 21-fold fewer answer-context memory tokens than the strongest non-oracle baseline.

S^2D-OPD Co-first author; under review at ICLR 2027

Direct-OPD transfers RL gains from weaker teachers to stronger students but supervises every state. For a collaborator's proposed method, I developed an alternative mathematical motivation, designed experiments, and wrote the paper. Retaining 10% of states by JSD matched or outperformed dense Direct-OPD across eight settings.

Research Experience

Shanghai AI Laboratory, LLM Center Dec. 2025 – Sep. 2026

Research Intern Mentor: Zichen Ding

Conducted the controlled study of reinforcement learning for search agents described above. Contributed to OS-Themis, a critic framework for GUI-agent rewards.

Academic Service and Languages

Reviewer: ICLR 2027; KDD 2027 (Cycle 1); COLM 2026; ARR (January 2026).
Teaching Assistant: Contemporary Artificial Intelligence (undergraduate), ECNU, Fall 2025.
English: IELTS, overall band 7.0.

Honors and Scholarships

National Scholarship (Graduate Students) 2025–2026

Bold Exploration Corporate Scholarship 2024–2025